

Requirements Traceability Matrix (RTM)

E-Commerce QA Portfolio Project

Requirement ID	Requirement Description	Test Case ID(s)
REQ-001	User can successfully log in using valid credentials	TC_LOGIN_01
REQ-002	User cannot log in with invalid credentials	TC_LOGIN_02, TC_LOGIN_03, TC_LOGIN_04
REQ-003	Email and password fields are required	TC_LOGIN_05, TC_LOGIN_06, TC_LOGIN_07
REQ-004	Email format validation is enforced	TC_LOGIN_08
REQ-005	Passwords are case-sensitive	TC_LOGIN_09
REQ-006	Remember Me functionality persists user credentials	TC_LOGIN_10
REQ-007	Account lockout occurs after multiple failed attempts	TC_LOGIN_11
REQ-008	Users can reset forgotten passwords	TC_LOGIN_12
REQ-009	Users can successfully log out	TC_LOGIN_13
REQ-010	User sessions expire after inactivity	TC_LOGIN_14
REQ-011	Protected pages require authentication	TC_LOGIN_15
REQ-012	Users can search for existing products	TC_SEARCH_01
REQ-013	System handles searches for non-existing products	TC_SEARCH_02
REQ-014	Partial keyword searches return relevant products	TC_SEARCH_03
REQ-015	Empty search submissions are handled appropriately	TC_SEARCH_04
REQ-016	Special character searches do not break functionality	TC_SEARCH_05
REQ-017	Search results are ranked by relevance	TC_SEARCH_06
REQ-018	Users can add products to the shopping cart	TC_CART_01

Requirement ID	Requirement Description	Test Case ID(s)
REQ-019	Users can remove products from the shopping cart	TC_CART_02
REQ-020	Users can update product quantities	TC_CART_03
REQ-021	Users can add multiple products to the cart	TC_CART_04
REQ-022	Cart contents persist after page refresh	TC_CART_05
REQ-023	Invalid cart quantities are rejected	TC_CART_06
REQ-024	Users can complete a successful checkout	TC_CHECKOUT_01
REQ-025	Shipping information is required for checkout	TC_CHECKOUT_02
REQ-026	Payment information must be valid	TC_CHECKOUT_03
REQ-027	Order confirmation information is accurate	TC_CHECKOUT_04
REQ-028	Users can cancel checkout before submission	TC_CHECKOUT_05

Defect Mapping

Defect ID	Related Requirement	Related Test Case
BUG_001	REQ-020	TC_CART_03
BUG_002	REQ-025	TC_CHECKOUT_02
BUG_003	REQ-020	TC_CART_03
BUG_004	REQ-009, REQ-011	TC_LOGIN_13, TC_LOGIN_15
BUG_005	REQ-017	TC_SEARCH_06

Total Requirements: 28

Total Test Cases: 32

Total Defects: 5

Coverage: 100% of documented requirements mapped to at least one test case.